

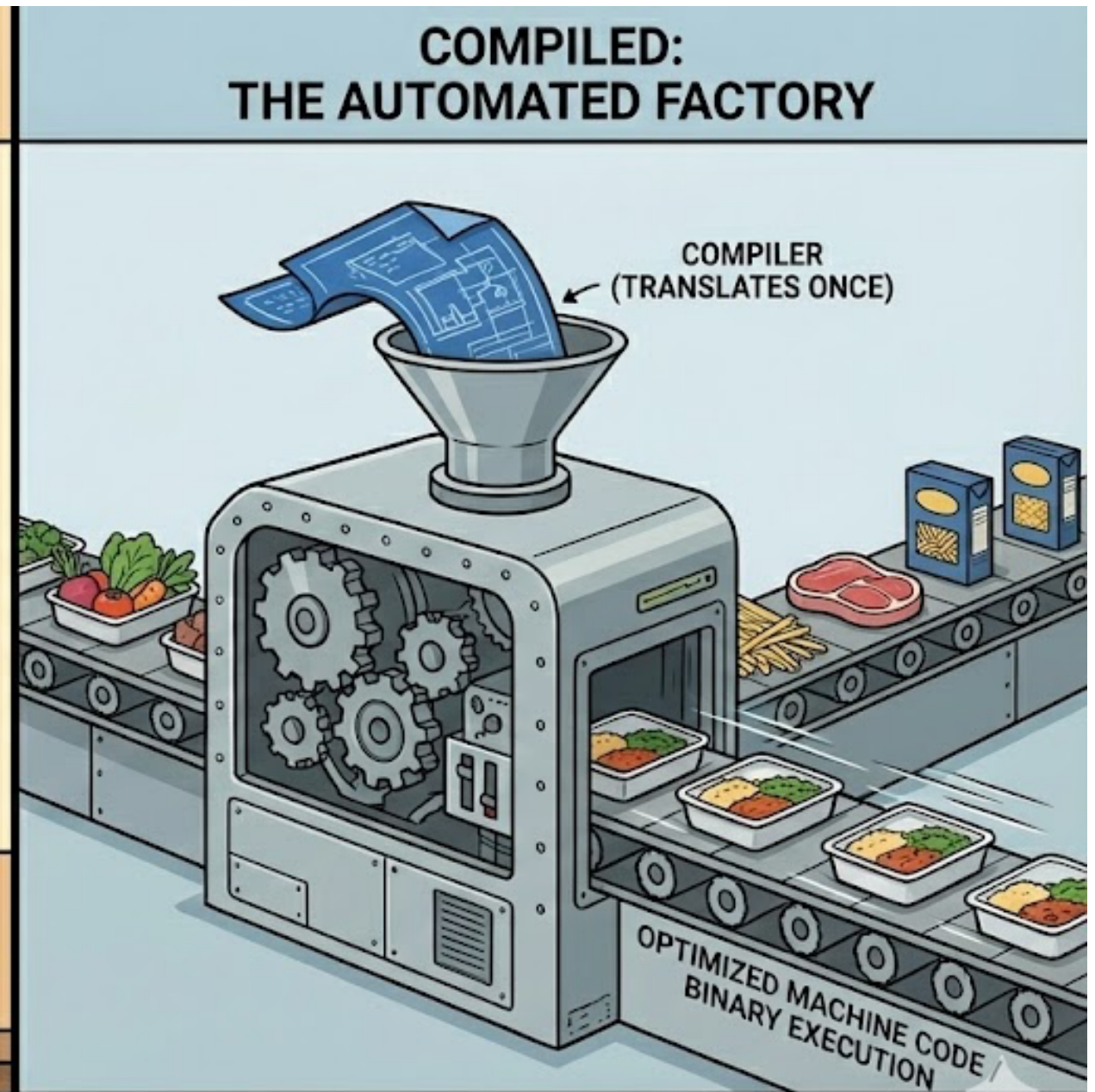
Lecture 14: Compiled languages

CEE 690

Compiled vs interpreted



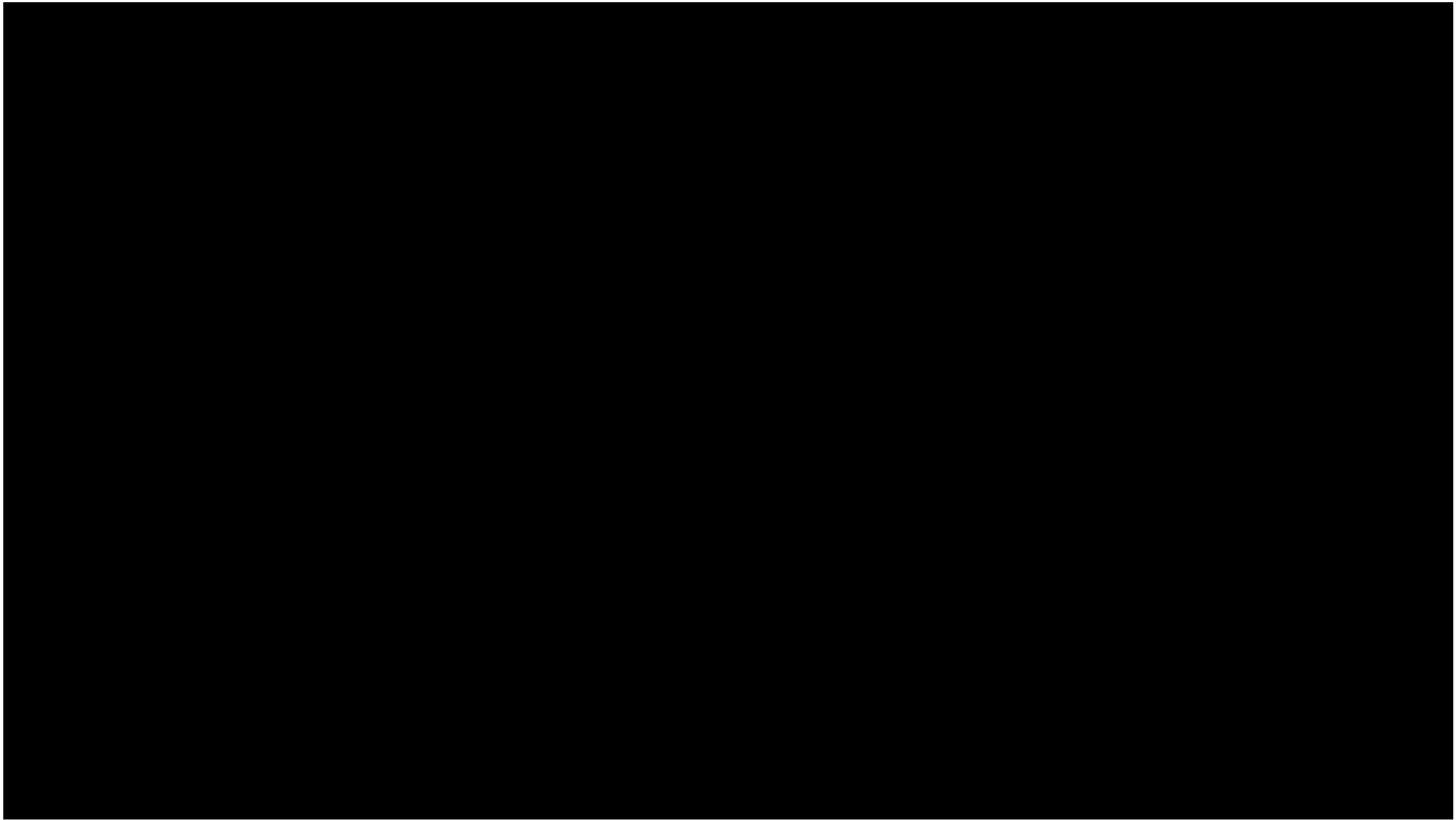
Flexible, Easy to Change, Slower Overhead



Rigid Setup, Upfront Cost, High Performance, Efficient

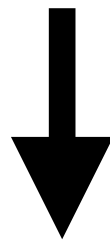
Building and running compiled code

1. **Source code (e.g., .c, .cpp and .f90)** - The human readable code that we create and modify.
2. **Compiler (e.g., gcc and fortran)** - The “machine” that translates the source code into a binary object.
3. **The executable (e.g., .out, .exe...)** - This the actual too. It is a collection of 0s and 1s that the CPU can execute without any further help (i.e., no need for an interpreter like Python or R)



Hello world in Fortran

```
program hello_world
    ! Comments in Fortran
    print *, "Hello, World"
end program hello_world
```



```
[nc153 $ gfortran -o hello_world.exe hello_world.f90
[nc153 $ ./hello_world.exe
Hello, World
```

(Anyone that is not in sciences/engineering will look very confused when you tell them that anyone is still using Fortran)